

Diffusion map based forecasting from eDNA abundance time series

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1

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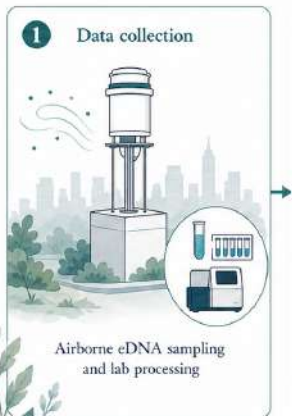
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Swedish
Biodiversity
In Time And
Space

A little overview

From raw eDNA samples to ecological forecasts: an overview



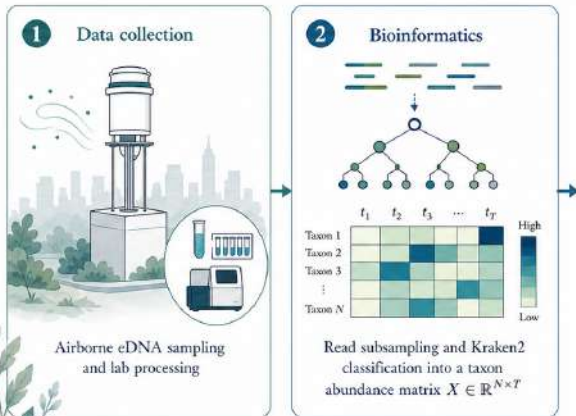
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From raw eDNA samples to ecological forecasts: an overview



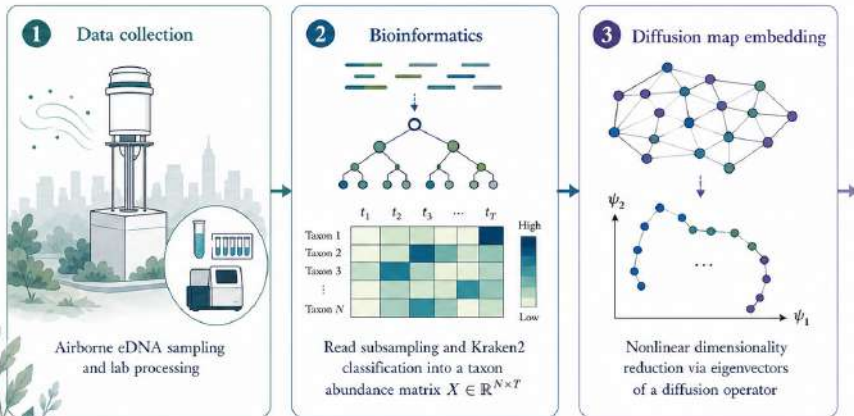
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From raw eDNA samples to ecological forecasts: an overview



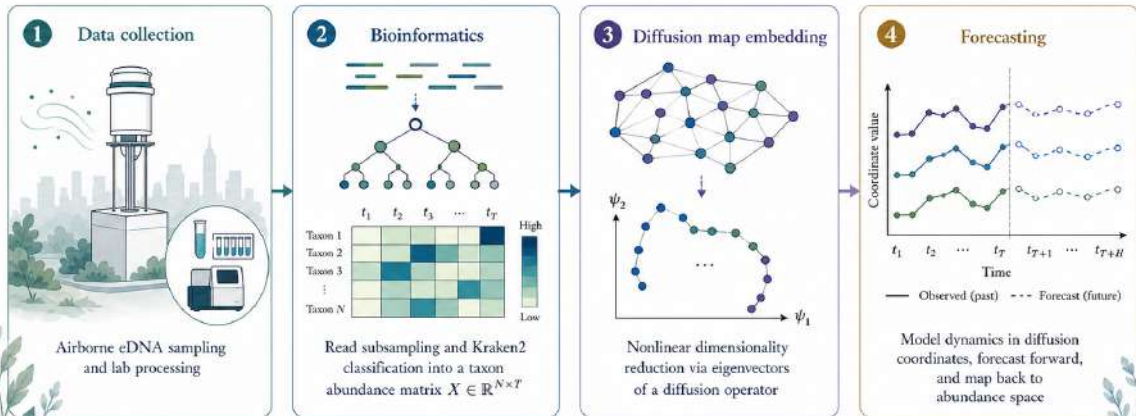
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From raw eDNA samples to ecological forecasts: an overview



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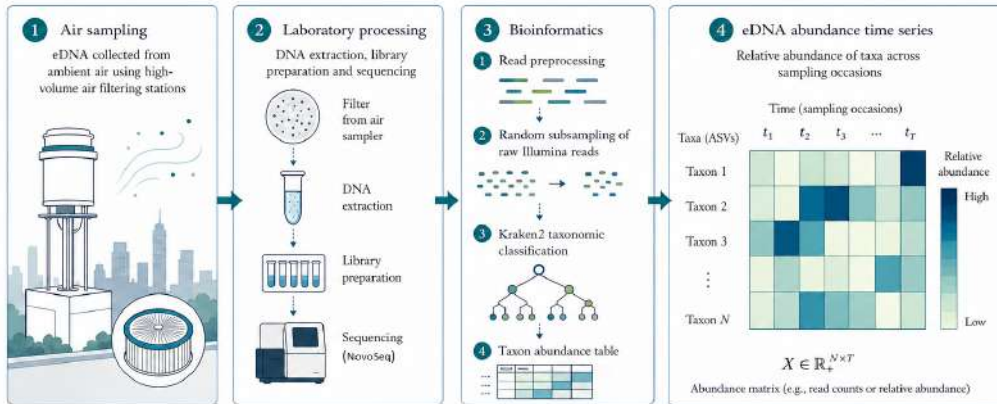
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Each step above is unpacked in detail in the figures that follow

Data collection and bioinformatics



Non-invasive monitoring
Species presence detected from environmental DNA in the air



High temporal resolution
Frequent sampling provides time series data



Broad biodiversity coverage
Detects multiple taxa across diverse ecosystems



Embedding

Related approaches

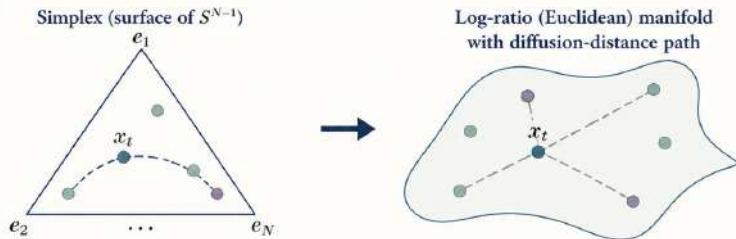
 Approach	 Core idea	 Strengths	 Limitations / relevance here
 EDM / Takens embedding	Reconstruct the attractor from time-delay coordinates of one or a few observed variables; forecast using local neighbors or local linear models (Simplex, S-Map).	Strong ecological pedigree; naturally nonlinear and state-dependent; interpretable local geometry.	Variable-specific and delay-based; struggles in very high-dimensional multi-taxon settings. Best viewed as complementary.
 Diffusion maps	Construct a kernel-based Markov operator on observations and use leading eigenvectors as low-dimensional coordinates preserving diffusion geometry.	Handles high-dimensional multivariate states; denoises; captures nonlinear manifold structure; gives a shared latent space for all taxa.	Requires kernel and normalization choices; risk of leakage if embedding uses the full time series. Motivates our Nyström out-of-sample construction.
 Deep learning	Learn latent temporal dynamics directly from data with flexible sequence models (e.g., RNNs, LSTMs, TCNs, Transformers).	Very flexible; strong predictive baselines when data are abundant; can model complex nonlinear temporal dependencies.	More data- and tuning-hungry; easy to introduce subtle leakage in preprocessing and validation; often harder to connect back to ecological state-space ideas.



Positioning of this talk: we combine the ecological state-space intuition of EDM with the shared low-dimensional geometry of diffusion maps, while keeping evaluation leakage-free and leaving deep models as important future baselines.

Why compositionality matters

The closure problem. Relative abundance vectors x_t in $\mathbb{R}_+^N$ live on the simplex ($\sum_i x_{t,i} = 1$), not in unconstrained Euclidean space. As emphasized in compositional data analysis (Aitchison, 1982), this closure constraint can mislead raw Euclidean distances and standard multivariate operations.



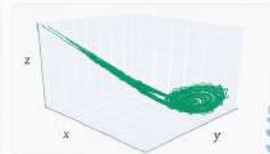
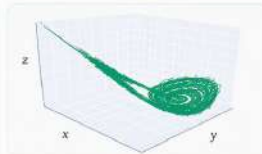
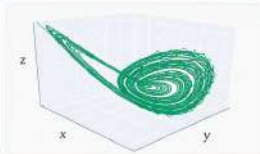
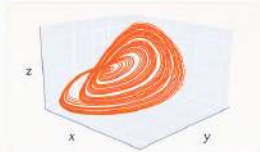
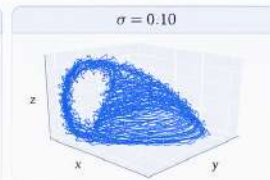
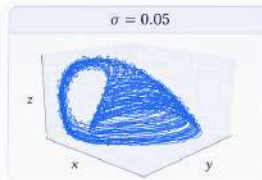
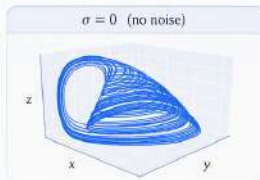
- **Closure induces spurious dependence.** Because compositions sum to one, if one component increases, at least one other must decrease—even if their absolute abundances are unchanged.
- **Log-ratio geometry.** Working in log-ratio coordinates (e.g., CLR or ILR) respects the simplex geometry and restores meaningful notions of distance and variation.
- **Our current choice.** We therefore map compositions to a real vector space using a log-ratio transform and use a compositional dissimilarity (e.g., Aitchison distance) for analysis, modeling, and forecasting.



Hastings-Powell - attractor degradation under Gaussian noise



Rows: Original / Absolute (closed) / CLR • Columns: noise $\sigma = 0, 0.05, 0.10 \times$ component std



Noise progressively degrades attractor geometry in raw and closed data.



Diffusion map approach

Diffusion Maps Pipeline

1 Data points in $\mathbb{R}^n$

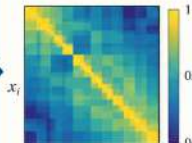


$$\mathcal{X} = \{x_1, \dots, x_N\} \subset \mathbb{R}^n$$

High-dimensional observations

2 Affinity kernel

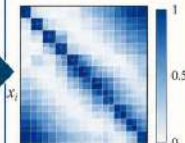
$$k(x, y) = \exp\left(-\frac{\|x - y\|^2}{\epsilon}\right)$$



$$K \in \mathbb{R}^{N \times N}, K_{ij} = k(x_i, x_j)$$

3 Markov matrix P

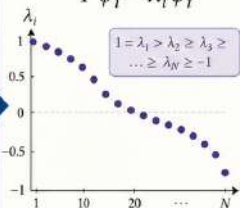
$$P = D^{-1}K$$



$$P \in \mathbb{R}^{N \times N}, P_{ij} = \frac{K_{ij}}{\sum_k K_{ik}}$$

4 Eigendecomposition

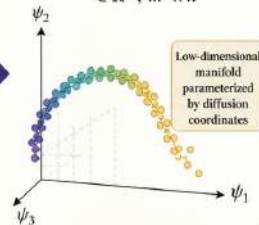
$$P\psi_i = \lambda_i\psi_i$$



Eigenvalues $\{\lambda_i\}$ (sorted)
Eigenvectors $\{\psi_i\}$ form an orthonormal basis

5 Diffusion coordinates

$$\Psi_t(x) = (\lambda_1^t \psi_1(x), \dots, \lambda_m^t \psi_m(x)) \in \mathbb{R}^m, m \ll n$$



Low-dimensional manifold parameterized by diffusion coordinates

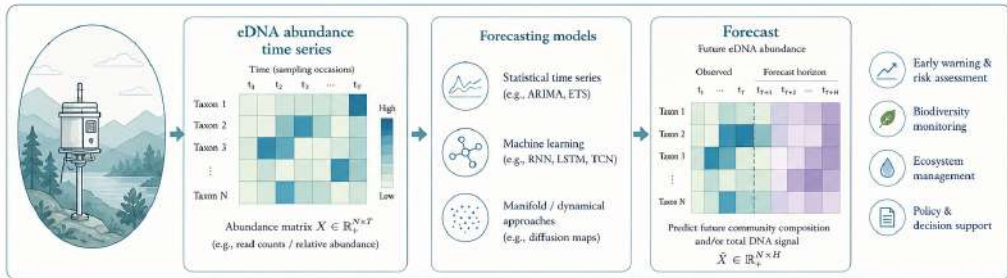


Low-dimensional, denoised representation that captures the intrinsic dynamics



forecast future eDNA abundance

Forecasting



Forecasting pitfalls



1. Ignoring seasonality and trends

Overlooking periodic patterns or long-term trends leads to systematic errors.

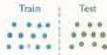
Example: Missing seasonal cycles in eDNA signals.



2. Overfitting

Model captures noise instead of signal, leading to poor out-of-sample performance.

Example: Too complex models with many parameters.



3. Data leakage

Information from the future unintentionally influences training, creating overly optimistic results.

Example: Using future information for scaling or feature creation.



4. Short or irregular time series

Too few time points or irregular sampling limits the ability to learn dynamics.

Example: Sparse or unequally spaced sampling.



5. Evaluating only in-sample fit

Good in-sample fit does not guarantee accurate future forecasts.

Example: High R^2 but poor forecast skill.



Best practices



Use time-aware validation (rolling origin / blocked CV).



Start simple, compare models, and report baselines.



Inspect residuals and forecast errors across horizons.



Incorporate domain knowledge (seasonality, trends, environmental drivers).

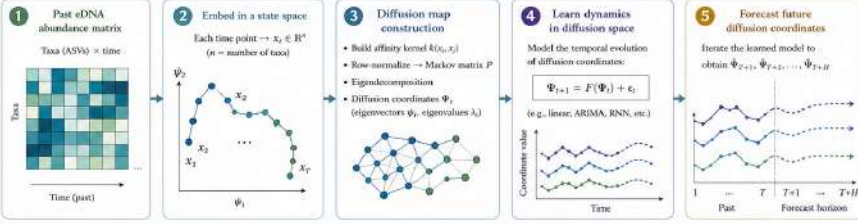


Ensure data quality, consistent preprocessing, and transparent reporting.



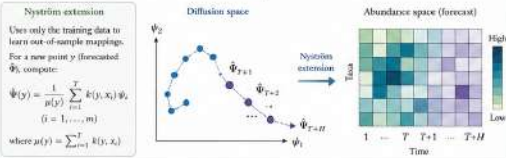
Robust forecasting of ecological communities requires careful data handling, appropriate models, and honest evaluation.

Diffusion map based forecasting: overview



6 Back-transform to abundance space (Nyström extension)

Map forecast diffusion coordinates back to abundance space without data leakage.



Key advantages*

- ✓ Eliminates data leakage via Nyström extension.
- ✓ Works with irregular sampling and high-dimensional data.
- ✓ Captures nonlinear ecological dynamics on a low-dimensional manifold.
- ✓ Scales to long-running monitoring time series.
- ✓ Provides probabilistic forecasts and uncertainty quantification (if using stochastic models).

*Assuming appropriate model validation and diagnostics.

From forecasted coordinates to ecological insight



Notation

$X \in \mathbb{R}^{N \times T}$: abundance matrix (N taxa, T past time points)
 $x_t \in \mathbb{R}^n$: state at time t (column of X)
 $\Psi_t \in \mathbb{R}^n$: diffusion coordinates at time t ($n \ll n$)
 H : forecast horizon (number of future steps)



Diffusion maps + Nyström extension enable leakage-free, data-driven forecasting of ecological communities from eDNA abundance time series.



Related work and key references

1



Coifman, R. R. and Lafon, S. (2006). Diffusion maps. *Applied and Computational Harmonic Analysis*, 21(1), 5-30.

Foundational spectral framework: kernels, Markov operators, and diffusion coordinates as manifold embeddings.

2



Wu, T., et al. (2024). Predicting multiple observations in complex systems through low-dimensional embeddings. *Nature Communications*, 15, 2242.

Feature-and-Reconstructed Manifold Mapping (FRMM): combining feature embedding and delay embedding for multi-variable forecasting.

3



Gault, C. M. et al. (2023). Dissimilarity analysis based on diffusion maps. *Oikos*, 2023(12), e10249.

Uses diffusion maps to analyze community data, applying bounded dissimilarities to recover gradients.

4



Sullivan, A. R., et al. (2025). Airborne eDNA captures three decades of ecosystem biodiversity. *Nature Communications*, 16, 11281.

34 years of weekly airborne eDNA revealing biodiversity dynamics.

Data example

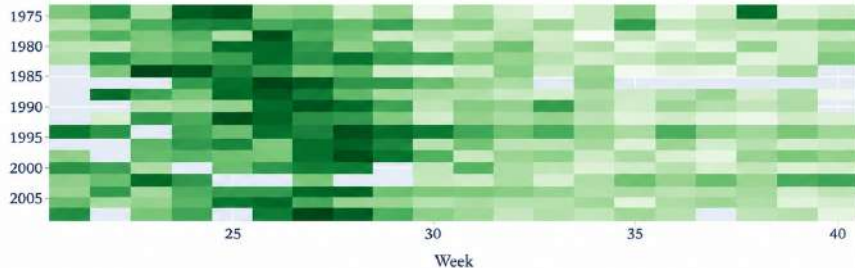
Example weekly abundance

Seasonal structure across observation years

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Year $\times$ week abundance profile

CLR abundance • darker green indicates higher relative abundance



Rows represent years; columns represent sampling weeks.

Illustrative abundance pattern

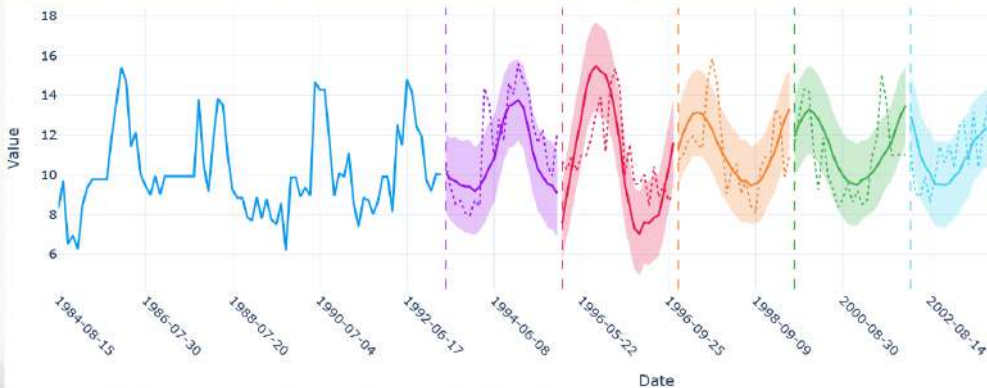
Example forecasting result

Historical signal, rolling forecast windows, and predictive uncertainty

1

Diffusion-map forecast across CV windows

Historical data, median forecasts, and 80% prediction intervals



Conclusions and future directions



Key takeaways



Airborne eDNA time series provide high-dimensional, long-running records of ecological communities



Diffusion maps recover a low-dimensional manifold that captures the intrinsic dynamics of community composition



Forecasting directly in diffusion coordinates, then mapping back via the Nystrom extension, avoids data leakage



This framework connects manifold learning (diffusion maps), empirical dynamic modeling, and ecological forecasting into one coherent pipeline



Future directions



Extend to multivariate, multi-site eDNA networks and cross-ecosystem comparisons



Develop early-warning indicators for regime shifts and biodiversity decline from diffusion coordinates



Combine diffusion maps with deep learning and generative diffusion models for hybrid forecasting



Scale to longer, higher-resolution monitoring datasets using landmark and out-of-sample diffusion map approximations



Diffusion maps offer a mathematically principled, leakage-free bridge between manifold learning and ecological forecasting





Link to swebits website